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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 09/10/2007 |
| **PI Name:** | Bacher, Jeff Ph.D. | | |
| **Project Title:** | A Novel Biodosimetry Method For Monitoring Radiation-Induced Genetic Damage | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Organization Name:** | Promega Corporation | | |
| **PI Address 1:** | Genetic Analysis | | |
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| **PI Web Page:** | | | |
| **City:** | Madison | **State:** | WI |
| **Zip Code:** | 53711-5399 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 02-OBPR-02 |
| **Start Date:** | 06/16/2003 | **End Date:** | 06/16/2007 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: no cost extension through 6/16/2007, per J. Dardano 2/06 | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Prolla, Thomas ( University of Wisconsin - Madison ) | | |
| **Grant/Contract No.:** | NAG9-1525 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | For extended space missions, exposure to hazardous radiation from galactic cosmic radiation and solar particle events poses a serious threat to the health of the astronauts. Dosimeters measure radiation dose received but not its biological effects or individual differences in radiation sensitivity. Accurate methods for measuring the extent of an individuals genetic damage resulting from accumulated radiation exposure is critical for estimating personal health risks and need for countermeasures. The overall goal of this proposal is to investigate the efficacy of a novel biodosimetry method for monitoring mutational load resulting from radiation-induced mutations in repetitive DNA sequences. Repetitive sequences constitute nearly one third of the human genome and some have been shown to be extremely sensitive to radiation-induced mutations. Most repetitive sequences are in non-coding regions of the genome and, therefore, alterations in these loci are usually not deleterious to the survival of the host cell. Thus, mutations in non-coding | | |

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| **Task Description:** | repetitive sequences can accumulate, providing a stable molecular record of DNA damage from all past exposures. Progenitor stem cells are long lived and continually divide producing identical duplicate progeny cells. Mutations in repetitive DNA loci in progenitor stem cells will accumulate and through analysis of progeny blood or sperm cells can serve as biological sensors to monitor radiation-induced genetic damage. To test this hypothesis we will (a) screen a broad range of highly mutable repetitive DNA sequences to identify genetic loci most sensitive to high-LET radiation-induced mutations, (b) show that radiation-induced mutations in progenitor stem cells are cumulative and that mutational load can be quantitatively measured by sampling a population of single or small pools of progeny blood or sperm cells for changes in repetitive DNA sequences, and (c) investigate possible mechanisms causing radiation-induced genomic instability in repetitive DNA loci. Knowledge obtained from completion of this proposal will determine the efficacy of a novel quantitative method for monitoring radiation-induced mutational load that would be applicable to both mouse models and humans. Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of radiation-induced mutations that would be useful in predicting and minimizing health risks. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Mutational load profiling, through analysis of mutations in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of mutations that may be useful for determining health risks and effectiveness of countermeasures for astronauts or other individuals exposed to ionizing radiation or chemical mutagens. Biomarkers identified in this study are also sensitive to free radical DNA damage and therefore may be useful markers for detection of cancer and other degenerative diseases in which oxidative stress is involved. |
| **Task Progress:** | Microsatellite repeat markers were identified that exhibited a linear dose response up to 1 Gy of 1 GeV/nucleon 56Fe ions and 137Cs gamma rays in mouse and human cells. Short tandem repeats on the Y chromosome and mononucleotide repeats on autosomal chromosomes exhibited significant increases in mutations at =0.5 Gy of 56Fe ions with frequencies averaging 4.3-10.3 x 10-3 mutations/locus, high enough for direct observation of mutations in irradiated cells. The sensitivity of the PCR-based microsatellite mutation assay was 10-fold greater than that observed for the SupFG1 transgenic mouse mutation assay. A significant increase in radiation-induced mutations in extended mononucleotide repeats was detectible in vivo in mouse blood and cheek samples 10 and 26 weeks after radiation exposure and these mutations were additive over multiple exposures. This study demonstrates the feasibility of a novel method for biodosimetry that is applicable to humans and other species. This new approach should complement existing methods of biodosimetry and might be useful for measuring radiation exposure in circumstances that are not amenable to current methods. |
| **Bibliography Type:** | Description: (Last Updated: 04/16/2019) |
| **Abstracts for Journals and Proceedings** | Bacher JW, Megid WA, Ensenberger MG, Halberg RB, Stanhope SA, Kent-First MG, Prolla TA, Storts D. "A Novel Biodosimetry Method." Presented at the 18th Annual NASA Space Radiation Investigators Workshop, San Francisco, CA, July 13-15, 2007.<br>18th Annual NASA Space Radiation Investigators Workshop, July 2007; page 145. , Jul-2007 |
| **Abstracts for Journals and Proceedings** | Bacher JW, Halberg RB, Megid WA, Ensenberger MG, Prolla TA. "A novel biodosimetry method for measuring radiation-induced genetic damage." Presented in a mini-symposium on biodosimetry at the 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5-8, 2006.<br>Radiation Research Society Abstracts Nov 2006; p.29. , Nov-2006 |
| **Abstracts for Journals and Proceedings** | Bacher JW, Megid WA, Ensenberger MG, Halberg RB, Stanhope SA, Kent-First MG, Prolla TA. "A Novel Method for Biodosimetry. " Presented at the 13th International Congress of Radiation Research, July 8-12, 2007.<br>International Congress of Radiation Research Abstract Book. 2007; p. 200. , Jul-2007 |
| **Articles in Peer-reviewed Journals** | Bacher JW, Sievers CK, Albrecht DM, Grimes IC, Weiss JM, Matkowskyj KA, Agni RM, Vyazunova I, Clipson L, Storts DR, Thliveris AT, Halberg RB. "Improved detection of microsatellite instability in early colorectal lesions." PLoS One. 2015 Aug 7;10(8):e0132727. eCollection 2015. https://doi.org/10.1371/journal.pone.0132727 ; PubMed PMID: 26252492; PubMed Central PMCID: PMC4529134 , Aug-2015 |
| **Articles in Peer-reviewed Journals** | Megid WA, Ensenberger MG, Halberg RB, Stanhope SA, Kent-First MG, Prolla TA, Bacher JW. "A novel method for biodosimetry." Radiat Environ Biophys. 2007 Jun;46(2):147-54. Epub 2006 Oct 27. PMID: 17072633 , Jun-2007 |